
Philosophical Transactions B Theme Issue "Getting it wrong about mistakes: the potential benefits of "misdirected" behaviour in animals"

From Jon Richardson <Jon.Richardson@glasgow.ac.uk>

Date Wed 1/21/2026 4:43 AM

To Nobuaki Mizumoto <nzm0095@auburn.edu>

Dear Authors,

We are delighted to inform you that our Theme Issue "**Getting it wrong about mistakes: the potential benefits of "misdirected" behaviour in animals**" has been approved for publication in *Philosophical Transactions B*. We look forward to working with you on this project.

In due course you will receive a separate email from the journal's Editorial Office with details of how to submit your paper and instructions about formatting it to the journal's requirements. However we wanted to take this opportunity to give you some specific instructions for this issue.

Please note that your paper will be required to fit the brief given when commissioned, and meet the high-quality standard of the journal. Therefore whilst we endeavour to allow opportunities to revise throughout the process, the Guest Editors and the journal's Editorial Board retain the right to reject papers. Prompt submission on schedule will allow for the maximum revision time possible.

Content

As a reminder here is our summary of the topic: *From Australian jewel beetles mating with beer bottles to turtles mistaking plastic bags for food, animals often seemingly misdirect their behaviour towards unsuitable targets. Misdirected behaviour occurs when fitness-related actions – like foraging, defence, mating, or parental care – are directed towards a target that provides no benefits and may even incur costs. Reproductive examples include directing sexual behaviour towards same-sex individuals, heterospecifics, close relatives, corpses, juveniles, or inanimate objects. Other behaviours that are often misdirected include foraging, aggression, egg-laying, and parental care. Misdirected behaviours are often dismissed as mistakes or anomalies. However, a more productive approach is to recognise that, in many contexts, the risk of misdirecting behaviour towards an unsuitable target is outweighed by the cost of failing to direct behaviour towards a suitable target. We call this having a "broad filter" – a framework derived from signal detection theory. Broader filters may be advantageous in many contexts and explain the widespread occurrence of misdirected behaviours. Re-examining misdirected behaviours through the lens of broad filters offers a more nuanced view of animal behaviour and challenges the assumption that misdirected behaviours are always mistakes. Better understanding misdirected behaviour is timely as these behaviours often involve animals responding to human-induced environmental changes and challenges.*

If you anticipate any changes from the information that was originally submitted following your invitation – including changes in co-authors – please let us know as soon as possible as some changes may need to be approved by the Editorial Office.

Schedule

As all papers in a particular issue publish together, publication can be significantly delayed by the last paper to be submitted. We want our issue to be timely and for papers to be published as quickly as possible, so prompt submission and quick turn-around of revisions will be appreciated.

Here is the planned schedule (please note this schedule has been shifted slightly later than the tentative schedule we initially proposed in your invitation). Please let us know if you are concerned about being able to meet any of

these dates.

Manuscripts submitted: **31st July 2026**

Peer review: August – October 2026

All revisions finalised: early 2027

Publication: early 2027

Length

The journal does not enforce a strict length limit, however papers that are overly long will need to be reduced. As a guide, please aim for no more than 9 printed pages. This equates to approximately 6000 words, 250 words of tables and 3 medium figures.

Data sharing

It is a condition of publication that authors make available the data and research materials supporting the results in the article. Datasets should be submitted as supplementary material or deposited in an appropriate, recognized repository and the associated accession number, link or DOI to the datasets must be included in the methods section of the article. See <https://royalsociety.org/journals/ethics-policies/> for more information.

Supplements

We encourage you to think about additions that could add extra value to your paper. This could be in the form of additional information for your paper (e.g. data, tables, figures, text, movies etc.) or more general things such as interviews with the authors, audio files from presentations etc.

Author list

For your interest we provide a full list of invited authors and their proposed titles below. Please be aware that this list is subject to change and is arranged in no particular order.

Thank you again for contributing to this issue and we look forward to working with you.

Jon Richardson (jon.richardson@glasgow.ac.uk)

Marlene Zuk (mzuk@umn.edu)

Mark Hauber (mhauber@gc.cuny.edu)

List of contributions

- 1. Intentional or mistaken? Termites start pairing broadly but refine choices**
Nobuaki Mizumoto
- 2. Demography, sexual conflict, and the evolution of sex discrimination**
Brian Lerch & Ewan Flinham
- 3. Nothing left to lose?: misdirected behaviour as a consequence of terminal investment**
Jon Richardson
- 4. Mis-shapes, mistakes, misfits: telling apart reproductive from non-reproductive sex**
Joanna Kennedy & David Shuker
- 5. Beyond Misidentification: Reciprocal Same-Sex Courtship in a Sexually Dimorphic Insect**
Terrence Chang & Andrew Mason
- 6. 'Grossest blunder' or best option? Mating behavior's role in hybridization**
Karin Pfennig
- 7. Genetic variation in susceptibility to an evolutionary trap in groups of fruit flies**
Lea Pollack & Julia Saltz
- 8. Mate-choice behaviour in invasive cane toads**
Richard Shine
- 9. The cost of flexibility: misdirected foraging and plastic ingestion in urban habitats**
Jânio Hermilindo Moreira de Mesquita, Igor Luis Kaefer, Daiani Kochhann

10. **Ancient mariners and plastic jellyfish: causes and consequences of marine debris ingestion in sea turtles**
Brendan Godley
11. **Balancing the costs of inbreeding and outbreeding in complex social environments**
EV (Ginny) Greenway
12. **Dappled light and decision-making in hosts of the brown-headed cowbird (*Molothrus ater*)**
Quentin Jamison, Silas Hernandez, Juliana Villa & Daniel Hanley
13. **Does costly care for brood parasites in the nest induce growth plasticity in the hosts' own offspring?**
Walter Svagelj, Jeanne Chaupitre, Jazmin Morel, Maia Lores, Abigail Scavarda, Mark Hauber, Vanina Fiorini
14. **The chemical basis of egg-mimicry: a multi-lineage review of the proximate basis of egg misrecognition in mimetic brood parasites of avian hosts**
Mark E. Hauber
15. **Adoption in spotted hyenas: a case of misplaced maternal investment?**
Marion East, Sarah Benhaïem, Sonja Metzger, Julius Nyahongo, Heribert Hofer
16. **Paradoxical contest decisions: resource holding potential beyond size advantage**
Mark Briffa
17. **Misguided responses to multiple dangers**
Andy Sih & Richard Szeligowski
18. **Accidental Affairs: Evolutionary Ecology of Misdirected Mating**
Geetanjali Mishra, Mohd Sariq, Mohnish Singh, Tripti Yadav
19. **Why we need ecology for a better understanding of animal decision-making**
John Bosu Mensah, Bárbara Freitas, Diogo Antunes, Max Ringler, Eva Ringler
20. **What can we learn from what animals get wrong?**
Jon Richardson, Marlene Zuk, Mark E. Hauber

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